

SANDEEP KUMAR

Machine Learning • IoT Engineering • Data Analytics

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PROFESSIONAL SUMMARY

Computer Science graduate specializing in IoT, Cybersecurity, and Blockchain, with hands-on experience building AI-powered and IoT systems — including an intrusion-detection dashboard and a wireless underground telemetry network. Track record of placing in three national and institute-level hackathons (1st Runner-Up at IHMMC 2026, 3rd Place at Initium 11.0, and SIH 2025 Waiting-List Finalist). Comfortable working across the full pipeline — from data preprocessing and model building to embedded hardware integration and live dashboards. Seeking a Machine Learning Engineer, Data Scientist, IoT Engineer, or Data Analyst fresher role to bring this cross-domain problem-solving to a production team.

SKILLS

Programming	Python, Java, C
Frontend Basics	HTML, CSS, JavaScript
ML & Data	Scikit-learn, Pandas, NumPy, Matplotlib, EDA, Feature Engineering, Classification, Regression
IoT & Embedded	Arduino UNO, ESP32, ESP8266, Sensor Integration, Real-Time Systems
Databases & BI	MySQL, SQL Server, Query Optimization, Power BI, Excel
Tools	Git, GitHub, VS Code, Jupyter Notebook, Google Collab, Flask

LANGUAGES

Hindi — Native | English — Professional Working Proficiency

PROJECTS

1. PreBreach Shield — AI-Based Intrusion Detection System

Team Project, Comex Hacks | Jun–Jul 2025

Link: github.com/Sandeep-Kumar-930/PreBreach-Shield-AI

- Built an ML-based intrusion detection system trained on network traffic data, covering preprocessing, feature engineering, and EDA to distinguish attack patterns from normal traffic.
- Integrated a Flask + MySQL backend with a live dashboard for real-time threat monitoring, taking the model from notebook to a working application.

Technologies: Python, Flask, MySQL, Pandas, NumPy, Matplotlib, Scikit-learn, Git, GitHub

2. VAYU-SETU — Wireless Telemetry Bridge for Underground Mine Safety

IEST Shibpur — 3rd Place | Mar 2026

Link: drive.google.com/file/

- Designed an internet-independent underground communication system using ESP32, ESP-NOW and LoRa(433 MHz), enabling real-time data transfer with no cellular or internet connectivity.
- Integrated gas, temperature, humidity, and water-level sensors with haptic alerts for real-time hazard detection to improve underground worker safety.

Technologies: ESP32, ESP-NOW, LoRa, Arduino IDE, Embedded C, Sensor Integration, Real-Time Monitoring

3. Sentinel-R-V4 — Multitasking Autonomous Rover

IHMMC 2026 — 1st Runner-Up | Mar 2026

Link: [\[working video\]](#)

- Co-developed an autonomous rover on Arduino UNO, ESP32, and Raspberry Pi 5 with sensor fusion (ultrasonic + IR + environmental) for obstacle detection and terrain navigation.
- Implemented wireless telemetry for remote control and live system monitoring, placing 1st Runner-Up out of competing teams.

Technologies: Arduino UNO, ESP32, Raspberry Pi 5, C/C++, Wireless Communication, Sensor Integration

4. GeoAdhikar—AI-Based FRA Monitoring System(SIH25108) Smart India Hackathon 2025, Ministry of Tribal Affairs | Sep–Oct 2025

Link: [\[github.com/Sandeep-Kumar-930/GeoAdhikar-SIH-2025\]](#)

- Contributed to an AI-powered WebGIS platform combining satellite imagery, NLP-based document digitization, and ML-based land classification for Forest Rights Act monitoring.
- Helped build a decision-support engine to guide central scheme allocation using spatial and socio-economic data.

Technologies: Python, NLP, WebGIS, GIS, Satellite Imagery, SQL, Data Analytics

5. Heart Disease Prediction Using ML

Semester 6 Group Project | Oct–Nov 2024

Link: [\[github.com/Sandeep-Kumar-930/Heart-disease-prediction\]](#)

- Built and compared Logistic Regression and Decision Tree models to predict heart disease risk from patient health data, including full preprocessing and EDA.
- Presented findings as a research paper at NCCCI 2025.

Technologies: Python, Scikit-learn, Pandas, NumPy, Matplotlib, Jupyter Notebook

ACHIEVEMENTS

- **1st Runner-Up — IHMMC 2026**, for Sentinel-R-V4 autonomous rover
- **3rd Place — Initium 11.0, IEST Shibpur**, for VAYU-SETU underground telemetry system
- **Smart India Hackathon (SIH) 2025** — Waiting-List Finalist, among top teams at Asansol Engineering College
- **Research Paper Presented** — "Heart Disease Prediction Using ML", NCCCI 2025
- **Top 60** — Google Cloud Gen AI Study Jams (twice)

EDUCATION

Asansol Engineering College (Maulana Abul Kalam Azad University of Technology)	2022 – 2026 (Expected)
B. Tech, Computer Science and Engineering (IoT, Cybersecurity & Blockchain)	CGPA: 7.18 / 10
Eastern Railway Inter College, Jamalpur, Bihar	2020 – 2022
Higher Secondary Education	65.4%
Eastern Railway Inter College, Jamalpur, Bihar	2019 – 2020
Secondary Education	69.6%

CERTIFICATIONS

- Google Cloud Gen AI Study Jams — Participant, ranked Top 60 (two cohorts)
- NCCCI 2025 — Research Paper Presentation, "Heart Disease Prediction Using ML"